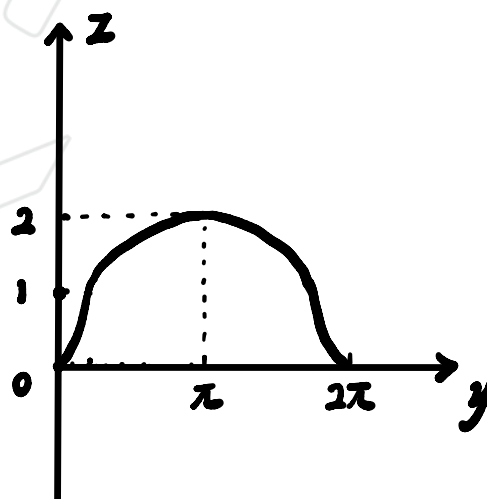
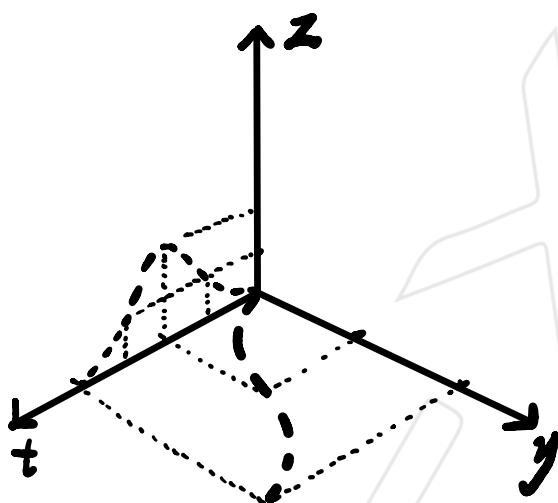
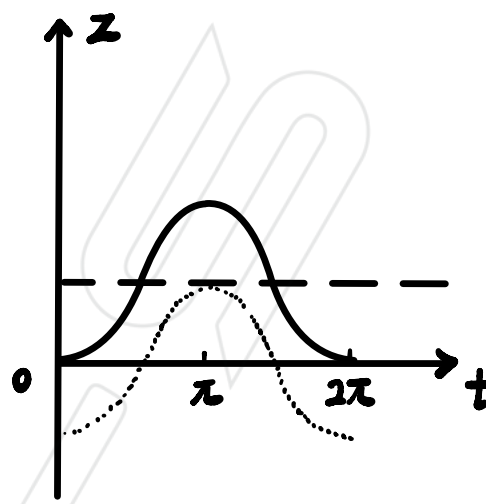
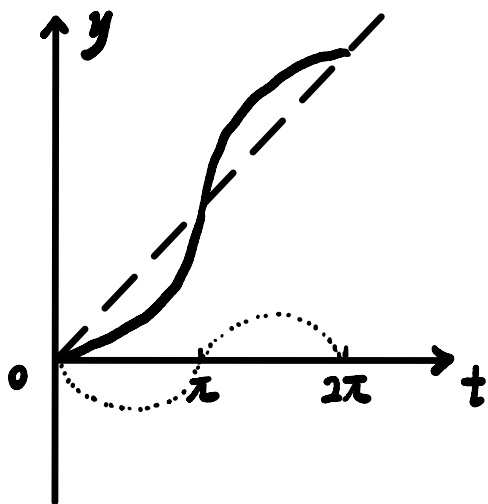


Problem set 3 - Path and line integrals

1. Consider a path given by $c(t) = (t - \sin t, 1 - \cos t)$ for $t \in [0, 2\pi]$. Sketch the path and find its arc length.



$$\begin{aligned}
 L &= \int_c ds = \int_0^{2\pi} \left\| \frac{dc}{dt} \right\| dt = \int_0^{2\pi} \sqrt{(1 - \cos t)^2 + (\sin t)^2} dt \\
 &= \int_0^{2\pi} (2 - 2\cos t)^{\frac{1}{2}} dt = \int_0^{2\pi} \sqrt{2} (1 - \cos t)^{\frac{1}{2}} dt \\
 &= \sqrt{2} \int_0^{2\pi} (2 \sin^2 \frac{1}{2} t)^{\frac{1}{2}} dt = 2 \int_0^{2\pi} \sin \frac{1}{2} t dt \\
 &= 2 [-2 \cos \frac{1}{2} t]_0^{2\pi} = 2(2 + 2) = 8
 \end{aligned}$$

2. A billiard ball on a square table follows the path $c(t) : [-1, 1] \rightarrow \mathbb{R}^3$ defined by $c(t) = (|t|, |t - \frac{1}{2}|, 0)$. Find the distance traveled by the ball (Hint: calculate the arc length of the path).

$$|f'(x)| = \frac{f'(x)}{|f'(x)|} f'(x)$$

$$\int_c ds = \int_{-1}^1 \left\| \frac{dc}{dt} \right\| dt$$

$$= \int_{-1}^1 \left\| \frac{t}{|t|}, \frac{t-\frac{1}{2}}{|t-\frac{1}{2}|}, 0 \right\| dt$$

$$= \int_{-1}^1 \sqrt{\frac{t^2}{t^2} + \frac{(t-\frac{1}{2})^2}{(t-\frac{1}{2})^2} + 0} dt$$

$$= \int_{-1}^1 \sqrt{2} dt$$

$$= [\sqrt{2}t]_{-1}^1$$

$$= 2\sqrt{2}$$

$$\text{OR } \int_c ds = \int_{-1}^0 \overset{(-t, -t+\frac{1}{2}, 0)}{\left\| \frac{dc}{dt} \right\|} dt + \int_0^{\frac{1}{2}} \overset{(t, -t+\frac{1}{2}, 0)}{\left\| \frac{dc}{dt} \right\|} dt + \int_{\frac{1}{2}}^1 \overset{(t, t-\frac{1}{2}, 0)}{\left\| \frac{dc}{dt} \right\|} dt$$

3. Consider a path given by a quarter-circle with the radius $r = 4$ in the first quadrant.

- (a) if the path is parameterised by $c(t) = (4 \cos t, 4 \sin t)$ for $0 \leq t \leq \pi/2$, evaluate $\int_c f ds$ where $f(x, y) = xy^2$.
- (b) Consider a reparametrisation of the path such that $c'(t') = (4 \cos 5t', -4 \sin 5t')$ with $-\pi/10 \leq t' \leq 0$. evaluate $\int_{c'} f ds$ and compare it with the answer in (a).

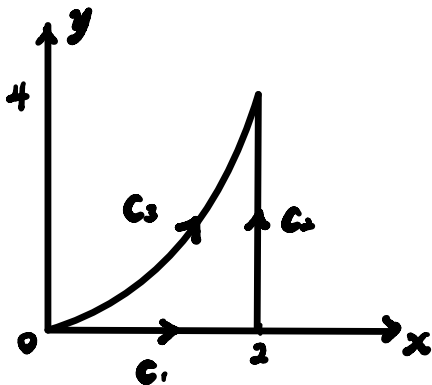
$$\begin{aligned}
 \text{(a)} \quad \int_c f ds &= \int_0^{\frac{\pi}{2}} xy^2 \left\| \frac{dc}{dt} \right\| dt \\
 &= \int_0^{\frac{\pi}{2}} 4 \cos t (4 \sin t)^2 \sqrt{(-4 \sin t)^2 + (4 \cos t)^2} dt \\
 &= \int_0^{\frac{\pi}{2}} 64 \sin^2 t \cos t \cdot 4 dt = \int_0^{\frac{\pi}{2}} 256 \sin^2 t \cos t dt \\
 \text{Let } u &= \sin t \quad \therefore \frac{du}{dt} = \cos t \\
 &= \int_0^1 256 u^2 du = \left[\frac{256}{3} u^3 \right]_0^1 = \frac{256}{3}
 \end{aligned}$$

$$\begin{aligned}
 \text{(b)} \quad \int_{c'} f ds &= \int_{-\frac{\pi}{10}}^0 xy^2 \left\| \frac{dc}{dt} \right\| dt \\
 &= \int_{-\frac{\pi}{10}}^0 4 \cos 5t' (-4 \sin 5t')^2 \sqrt{(4 \cos 5t')^2 + (-4 \sin 5t')^2} dt' \\
 &= \int_{-\frac{\pi}{10}}^0 256 \sin^2 5t' \cos 5t' \cdot 5 dt' \\
 \text{Let } u &= \sin 5t' \quad \therefore \frac{du}{dt} = 5 \cos 5t' \\
 &= \int_{-1}^0 \frac{256}{5} u^2 du = \left[\frac{256}{15} u^3 \right]_{-1}^0 = \frac{256}{15} \\
 &= \frac{1}{5} \int_c f ds
 \end{aligned}$$

4. Evaluate the integral $\int_C \cos z \, dx + e^x \, dy + e^y \, dz$, where $c(t) = (1, t, e^t)$ for $0 \leq t \leq 2$.

$$\begin{aligned}
 \int_C F \cdot ds &= \int_C (\cos z, e^x, e^y) \cdot ds \\
 &= \int_C (\cos e^t, e^1, e^t) \cdot (0, 1, e^t) \, dt \\
 &= \int_0^2 e + e^{2t} \, dt \\
 &= [et + \tfrac{1}{2}e^{2t}]_0^2 \\
 &= 2e + \tfrac{1}{2}e^4 - (0 + \tfrac{1}{2}) \\
 &= \tfrac{1}{2}e^4 + 2e - \tfrac{1}{2}
 \end{aligned}$$

5. Let c be a simple closed curve composed of the following three curves: $y = 0$, $x = 2$, and $y = x^2$. Evaluate $\int_c \mathbf{F} \cdot d\mathbf{s}$, where $\mathbf{F}(x, y) = (y^2, -x^2)$.



$$C_1: (t, 0) \text{ for } t \in [0, 2]$$

$$C_2: (2, t) \text{ for } t \in [0, 4]$$

$$C_3: (t_3, t_3^2) \text{ for } t_3 \in [0, 2]$$

$$\nabla \times \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ y^2 & -x^2 & 0 \end{vmatrix} = 0\mathbf{i} + 0\mathbf{j} + (-2x - 2y)\mathbf{k}$$

$$\int_c \mathbf{F} \cdot d\mathbf{s} = \int_{C_1} \mathbf{F} \cdot d\mathbf{s} + \int_{C_2} \mathbf{F} \cdot d\mathbf{s} - \int_{C_3} \mathbf{F} \cdot d\mathbf{s}$$

$$= \int_0^2 (0, -t^2) \cdot (1, 0) dt$$

$$+ \int_0^4 (t^2, 0) \cdot (0, 1) dt$$

$$- \int_0^2 (t_3^4, -t_3^2) \cdot (1, 2t_3) dt$$

$$= \int_0^2 -t^2 dt + \int_0^4 t^2 dt - \int_0^2 (t_3^4 - 2t_3^3) dt = \left[-\frac{1}{3}t^3 + \frac{1}{3}t^3 - \frac{1}{2}t^4 \right]_0^2$$

$$= -\frac{8}{3} + 16 = \frac{40}{3}$$

6. Let c be the path defined by $c(\theta) = (\cos^3 \theta, \sin^3 \theta, \theta)$ for $0 \leq \theta \leq 7\pi/2$. If a vector field $F : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ is given by $F = (\sin z, \cos z, -(xy)^{1/3})$, evaluate $\int_c F \cdot ds$.

$$\int_c F \cdot ds$$

$$= \int_0^{7\pi/2} (\sin z, \cos z, -(xy)^{1/3}) \cdot (-3\sin\theta \cos^2\theta, 3\cos\theta \sin^2\theta, 1) d\theta$$

$$= \int_0^{7\pi/2} (\sin\theta, \cos\theta, -\cos\theta \sin\theta) \cdot (-3\sin\theta \cos^2\theta, 3\cos\theta \sin^2\theta, 1) d\theta$$

$$= \int_0^{7\pi/2} (-3\sin^2\theta \cos^2\theta + 3\cos^2\theta \sin^2\theta - \cos\theta \sin\theta) d\theta$$

$$= \int_0^{7\pi/2} -\cos\theta \sin\theta d\theta$$

$$\text{Let } u = \cos\theta, \quad \frac{du}{d\theta} = -\sin\theta$$

$$= \int_1^0 u du$$

$$= \left[\frac{1}{2}u^2 \right]_1^0 = -\frac{1}{2}$$

7. Check if the following vector field is a conservative field (or irrotational). If it is, calculate its scalar potential.

(a) $F(x, y) = (e^{xy}, e^{x+y}),$

(b) $G(x, y) = (2x \cos y, -x^2 \sin y).$

(a)

$$\nabla \times F = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ e^{xy} & e^{x+y} & 0 \end{vmatrix}$$

$$= 0\mathbf{i} + 0\mathbf{j} + (e^{x+y} - xe^{xy})\mathbf{k} \neq 0$$

$\therefore F$ is rotational

7. Check if the following vector field is a conservative field (or irrotational). If it is, calculate its scalar potential.

(a) $F(x, y) = (e^{xy}, e^{x+y})$,

(b) $G(x, y) = (2x \cos y, -x^2 \sin y)$.

(b)

$$\nabla \times G = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 2x \cos y & -x^2 \sin y & 0 \end{vmatrix}$$

$$= 0\mathbf{i} + 0\mathbf{j} + (-2x \sin y + 2x \sin y)\mathbf{k} = 0$$

$\therefore G$ is irrotational

$$\therefore G = \nabla f(x, y)$$

$$\therefore f_x = x^2 \cos y + h(x) + C$$

$$f_y = x^2 \cos y + h(y) + C$$

$$\therefore f = x^2 \cos y + C$$

8. A mass M at the origin in $\mathbb{R}^3$ (i.e. $(x, y, z) = (0, 0, 0)$) exerts a force on a mass m located at $\mathbf{r} = (x, y, z)$ with magnitude GmM/r^2 and directed towards the origin (G is the gravitational constant and $r = \|\mathbf{r}\|$). In such a circumstance, the force field is given by

$$\mathbf{F}(x, y, z) = -\frac{GmM\mathbf{r}}{r^3}. \quad (1)$$

(a) Is $\mathbf{F}$ a conservative field?

(b) If yes, show that $\mathbf{F} = -\nabla\phi$ if $\phi(x, y, z) = -GmM/r$.

$$(a) \mathbf{F}(x, y, z) = \left(-\frac{GmMx}{r^3}, -\frac{GmMy}{r^3}, -\frac{GmMz}{r^3} \right)$$

$$\nabla \times \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ -\frac{GmMx}{r^3} & -\frac{GmMy}{r^3} & -\frac{GmMz}{r^3} \end{vmatrix}$$

$$= 0\mathbf{i} + 0\mathbf{j} + 0\mathbf{k} = 0$$

$\therefore \mathbf{F}$ is a conservative field

$$(b) \mathbf{F} = \nabla f(x, y, z)$$

$$f_x = -\frac{GmMx^2}{2r^3} + h(y, z) + C$$

$$f_y = -\frac{GmMy^2}{2r^3} + h(x, z) + C$$

$$f_z = -\frac{GmMz^2}{2r^3} + h(x, y) + C$$

$$\begin{aligned}
 \therefore f &= -\frac{GmMx^2}{2r^3} - \frac{GmMy^2}{2r^3} - \frac{GmMz^2}{2r^3} + C \\
 &= -\frac{GmM}{2r} \left(\frac{x^2 + y^2 + z^2}{x^2 + y^2 + z^2} \right) + C \\
 &= -\frac{GmM}{2r} + C
 \end{aligned}$$

$$\nabla\left(\frac{1}{r}\right) = -\frac{\vec{r}}{r^3}$$

$$\begin{aligned}
 \therefore -\nabla\phi &= -\nabla\left(\frac{GmM}{r}\right), \\
 &= -GmM \cdot \nabla\left(\frac{1}{r}\right) = GmM \cdot \frac{\vec{r}}{r^3} \\
 &= \vec{F}
 \end{aligned}$$